

MARLA, Australia

WMO: 944770

Lat: 27.300S

Lon: 133.620E

Elev: 1063

StdP: 14.14

Time Zone: 9.50 (AUA)

Period: 94-05

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	36.0	38.6	14.5	11.9	70.4	17.7	14.0	71.9	21.1	58.9	17.5	66.6	0.3	310	0.377

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	23.4	106.7	67.6	104.1	66.8	101.4	65.9	74.5	87.3	72.9	88.6	71.0	89.5	6.8	50

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
71.5	120.8	79.8	68.5	109.0	79.3	65.3	97.2	78.7	38.8	86.7	37.3	89.1	35.5	89.3	82.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 19.9	16.8	14.4	30.7	111.3	2.6	1.8	28.8	112.6	27.3	113.7	25.8	114.7	23.9	116.1		
(5)			27.9	77.5	3.4	3.1	25.5	79.8	23.5	81.6	21.6	83.4	19.2	85.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	70.1	86.1	84.9	77.9	68.4	60.8	54.8	54.2	57.4	67.1	71.2	77.6	82.2
	DBStd	13.03	6.72	7.06	7.06	6.05	6.73	5.98	5.22	6.07	7.27	8.01	7.78	7.18
	HDD50	33	0	0	0	0	0	14	14	5	0	0	0	0
	HDD65	1190	0	0	2	29	170	316	336	249	60	25	3	0
	CDD50	7387	1119	977	866	553	337	157	146	234	514	657	827	1000
	CDD65	3066	654	557	403	131	41	9	2	13	123	217	381	535
	CDH74	42001	9768	8211	5039	1487	416	72	40	211	1553	2790	4882	7532
	CDH80	23767	6225	5108	2617	504	101	13	3	43	650	1348	2660	4495
Wind	WSAvg	5.1	6.1	5.8	5.0	4.1	3.6	3.2	3.9	4.8	5.8	6.4	6.2	6.0
Precipitation	PrecAvg	9.3	0.7	1.4	0.5	0.5	0.6	0.6	0.4	0.4	0.5	0.8	0.9	1.3
	PrecMax	15.7	3.5	7.1	2.5	3.3	2.5	3.2	3.4	2.5	2.1	2.9	3.1	5.0
	PrecMin	4.2	0.1	0.0	0.0	0.0	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0
	PrecStd	3.7	0.9	1.8	0.7	0.8	0.7	0.7	0.7	0.6	0.6	0.9	0.8	1.3
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	109.7	109.2	103.0	94.5	87.6	82.5	79.0	86.3	96.9	102.4	106.7	107.3
		MCWB	68.9	70.2	67.6	62.5	60.2	59.2	55.2	58.9	61.6	62.3	65.1	65.9
	2%	DB	106.9	105.7	98.5	89.3	83.0	76.5	74.5	80.3	92.5	96.9	101.3	104.2
		MCWB	68.5	67.6	67.2	60.7	59.2	57.5	54.1	55.9	59.9	62.8	64.6	65.9
	5%	DB	104.1	103.1	95.8	86.0	79.2	71.6	70.8	75.9	88.2	92.3	97.6	100.8
		MCWB	67.5	66.8	66.1	59.1	57.8	54.8	52.9	54.5	59.3	60.8	63.3	64.7
	10%	DB	100.8	100.0	92.2	82.6	75.3	67.9	67.1	71.6	83.1	87.5	93.5	97.7
		MCWB	66.2	67.0	63.7	59.2	56.4	53.3	51.1	52.1	57.5	59.4	62.5	64.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	76.2	76.7	73.5	69.1	62.9	62.7	62.1	61.2	64.7	67.9	70.8	74.2
		MCDB	86.5	86.2	88.0	79.0	82.1	74.3	70.2	79.7	85.8	88.3	87.1	88.0
	2%	WB	74.4	74.9	71.2	66.7	60.7	60.2	57.8	58.1	62.9	65.6	68.3	71.6
		MCDB	84.4	88.9	88.6	75.5	78.8	70.7	67.3	75.4	81.7	85.5	85.5	88.1
	5%	WB	71.9	73.6	69.3	63.6	59.2	57.5	54.7	55.8	60.6	63.0	67.1	69.5
		MCDB	91.1	89.8	87.3	78.2	74.3	70.0	67.6	71.7	82.5	82.2	86.4	86.5
	10%	WB	69.6	71.8	67.0	61.6	57.9	54.8	52.3	53.7	59.0	61.2	65.7	67.4
		MCDB	93.3	88.9	86.4	77.8	72.3	65.4	64.7	67.9	81.0	82.1	85.9	87.3
Mean Daily Temperature Range	5% DB	MDBR	23.4	22.5	22.8	24.0	23.2	22.9	23.6	25.1	25.4	24.7	23.7	23.1
		MCDBR	25.6	26.2	24.9	28.5	26.8	26.8	29.6	29.6	31.4	29.9	28.9	26.8
		MCWBR	8.4	8.4	8.5	11.7	12.0	13.1	15.3	14.0	11.6	11.4	9.7	8.8
		MCDDBR	19.1	17.3	19.2	19.9	22.1	19.9	22.7	25.4	27.4	23.6	23.3	21.2
	5% WB	MCWBR	6.6	7.3	7.6	8.9	10.6	11.0	14.0	12.3	11.1	10.5	7.7	8.6
Clear-Sky Solar Irradiance	taub	0.362	0.357	0.337	0.322	0.297	0.283	0.286	0.300	0.334	0.348	0.360	0.353	
	taud	2.478	2.519	2.542	2.562	2.595	2.617	2.592	2.521	2.434	2.422	2.442	2.495	
	Ebn at Noon	311	307	304	294	288	287	290	297	300	306	309	314	
	Edh at Noon	37	35	32	29	25	23	25	30	36	38	38	37	
All-Sky Solar Radiation	RadAvg	2524	2337	2056	1643	1316	1146	1243	1579	1951	2235	2399	2472	
	RadStd	121	188	128	105	100	79	76	76	95	160	125	140	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (2 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page